Bioinformatics, 2026, pp. 1–11

doi:
Original Paper

ORIGINAL PAPER

Five Axes of Fragility: A Systematic Robustness Audit of Gene-Regulatory-Network Inference from Single-Cell Foundation Models

Ihor Kendiukhov[1,*]

[1]Department of Computer Science, University of Tübingen, Germany
[*]Corresponding author. ihor.kendiukhov@student.uni-tuebingen.de
FOR PUBLISHER ONLY Received on Date Month Year; revised on Date Month Year; accepted on Date Month Year

Abstract

Motivation: Gene-regulatory-network (GRN) inference from single-cell transcriptomics is increasingly used to prioritize regulatory hypotheses, yet most benchmarks evaluate accuracy under fixed preprocessing without testing whether rankings survive realistic analytical perturbations, and most published fragility claims are made under a single scorer family.
Results: We audit GRN edge rankings across five fragility axes—cell count, clustering resolution, rare-cell reliability, donor generalization, and integration sensitivity—on four Tabula Sapiens tissues, one non-Tabula-Sapiens external lung cohort (SmartSeq2; Travaglini et al., 2020), and one disease cohort (Delorey et al. 2021 COVID-19 lung atlas, 116,313 cells, 27 donors), using five scorer families: Pearson correlation, mutual information, GRNBoost2, GENIE3, and scGPT attention. Our central findings are that: (i) cluster-resolution stability under a bounded-drift Resolution-Sensitivity-Score sits 9–13 standard deviations below a rank-permutation null, yet every cohort fails a within-coarse constrained shuffle, so dual-null calibration recommends coarse or hybrid clustering across the full 66-point weight simplex; (ii) the rare-vs-abundant reliability-gate gap direction is preserved at 100% of 5,880 reasonable threshold combinations, establishing that the 0% rare passage rate is not an artifact of threshold choice; (iii) the minimum valid cell count (MVCC) is strongly scorer-dependent: under scGPT attention it exceeds 3,000 cells, whereas under Pearson-over-highly-variable-genes it is 200 cells for four of six cohorts and 500 cells for the two largest ones (primary immune and Delorey COVID lung), and it is anchor-invariant; (iv) 6–8 donors are needed for immune generalization under strict cells-per-donor balancing, and fewer under looser balancing; (v) integration method choice is not neutral: Harmony preserves rankings reasonably (Spearman 0.63–0.87, sign-flip 6–17%), scVI moderately perturbs them (Spearman 0.36–0.57, sign-flip 18–30%), and Scanorama on kidney and lung produces near-random rankings (Spearman ≈ 0, sign-flip 44–47%); (vi) scGPT attention rankings are statistically uncorrelated with classical-scorer rankings on matched kidney cells (Spearman $\rho \leq 0.20$, top-100 Jaccard ≤ 0.06), so within-scorer fragility profiles do not transfer across scorer families. A degree-preserving edge-rewire null is cleared at $p = 0.02$ on every cohort, rebutting the possibility that the within-coarse-shuffle failures reflect null over-stringency rather than genuine signal loss. No single cohort passes all five axes under calibrated criteria.
Availability and Implementation: The unified fragility package, reproducing every CSV and figure from raw h5ad, is at `https://github.com/Biodyn-AI/grn-robustness-audit` and archived at `https://doi.org/10.5281/zenodo.20009065`.
Contact: ihor.kendiukhov@student.uni-tuebingen.de
Supplementary Information: Supplementary figures (full Jaccard-vs-cell-count curves, weight-simplex sweep, 5,880-point threshold-sensitivity heatmap, triple-null calibration, gene-dimension and normalization ablations) are available at *Bioinformatics* online.

Key words: gene regulatory networks, robustness, single-cell transcriptomics, benchmarking, batch correction, rare cell types, reproducibility

Introduction

Single-cell RNA-sequencing enables transcriptome-wide inference of gene regulatory networks (GRNs). Classical correlation- and tree-based scorers (SCENIC (Aibar et al., 2017), GRNBoost2 (Moerman et al., 2019), GENIE3 (Huynh-Thu et al., 2010), ARACNe-family methods (Margolin et al., 2006)) and attention-derived scores from single-cell foundation models such as scGPT (Cui et al., 2024) and Geneformer (Theodoris et al., 2023) all rank TF–target edges as regulatory hypotheses, but confidence in those rankings depends on analytical choices that are rarely stress-tested. Published benchmarks (Pratapa

© The Author 2026. Published by Oxford University Press. All rights reserved. For permissions, please e-mail: journals.permissions@oup.com

et al., 2020; Chen and Mar, 2018) evaluate accuracy against fixed references under fixed preprocessing, leaving open the question of whether the same edges survive a different cell count, clustering resolution, integration method, or donor cohort. Atlas-level integration benchmarks (Luecken et al., 2022; Tran et al., 2020) document that batch correction perturbs downstream analyses, but the specific impact on TF–target ranking stability has not been systematically characterised.

We address this gap by conducting a systematic multi-axis robustness audit of TF–target edge rankings across five scorer families (Pearson correlation, mutual information, GRNBoost2, GENIE3, and scGPT attention) and six cohorts (four Tabula Sapiens tissues, one external-lab lung dataset, and one COVID-19 disease atlas). Our audit spans five axes that correspond to the most common sources of analytical variation in single-cell GRN workflows:

1. **Cell-count subsampling:** How many cells are needed for stable rankings, is that threshold dependent on the reference-anchor cell count, and do prior-constrained policies rescue small-sample regimes?
2. **Cluster-resolution sensitivity:** Does switching between coarse and fine clustering change which edges are prioritized, and does the conclusion survive rank-permutation nulls, weight-choice sensitivity, gene-dimension sweeps, and a triple-null calibration?
3. **Rare-cell-type reliability:** Do rare cell types yield mechanistic edges that pass the same confidence gates as abundant types, and how robust is that gap to the choice of gate thresholds?
4. **Donor generalization:** How many donors are needed for rankings to transfer to held-out individuals, and is the threshold dependent on cells-per-donor balancing choices?
5. **Integration-method sensitivity:** Do batch-correction methods (Harmony, scVI, and Scanorama) alter edge rankings, and do any of those changes improve biological validity as measured by perturbation-backed evidence rather than literature precision alone?

Beyond the per-axis questions, we ask an orthogonal one: *do fragility conclusions drawn under one scorer transfer to another?* We find that scGPT attention rankings are statistically uncorrelated with Pearson / mutual information / GRNBoost2 / GENIE3 rankings on the same cells and the same edge universe ($\rho \leq 0.20$, top-100 Jaccard ≤ 0.06), which generalises the message from "rankings are fragile under preprocessing perturbations" to "rankings are fragile under preprocessing perturbations *and* scorer choice."

Each axis was developed through an adversarial-review loop in which an AI-assisted review agent systematically challenged the initial analysis for methodological weaknesses (confounded null designs, overfitted hyperparameters, hidden data leakage, uninterpretable composite metrics, panel-selection arbitrariness), corrective experiments were designed and executed, and claims were updated accordingly. A full protocol description (base model, prompt templates, round-by-round logs, and a pre-registered audit of 27 agent-raised issues under a frozen rubric) is provided in Supplementary Methods S2.

The remainder of this paper describes the shared experimental setup (Section 2), presents per-axis results (Section 3), reports a scorer-divergence analysis (Section 3.6), analyses cross-axis interactions (Section 3.7), and provides a practical guidelines table for practitioners (Section 4).

Methods

Datasets, preprocessing, and panels

All analyses used processed single-cell datasets from the Tabula Sapiens atlas (The Tabula Sapiens Consortium, 2022), one external lung cohort (Travaglini et al. 2020 SmartSeq2), and one COVID-19 disease lung atlas (Delorey et al. 2021 (Delorey et al., 2021); Table 1). Datasets were preprocessed using Scanpy (Wolf et al., 2018) following established best practices (Luecken and Theis, 2019; Heumos et al., 2023). The primary normalization was depth-normalization ("`normalize_total`" with "`target_sum`=" 10^4) followed by `log1p`, with the top 600 highly-variable genes selected using the Seurat v3 flavor (Stuart et al., 2019). Two alternative normalization schemes—median-ratio size-factor normalization and Pearson residuals (Lause et al., 2021)—were evaluated as an ablation (Supplementary Fig. S6) and produced the same calibrated recommendation on every axis we examined. Every pipeline run stamps the full preprocessing parameter set into its provenance record; numerical reproducibility is therefore traceable to an explicit configuration file rather than to narrative prose.

TF–target edge panels used in axes with panel-restricted scoring were loaded from a shared registry (Supplementary Table S1) that resolves each panel to an exact edge list at runtime. Named panels include: "*primary*" (DoRothEA (Garcia-Alonso et al., 2019) confidence A/B ∩ TRRUST (Han et al., 2018), 3,384 edges before gene-universe restriction); "*dorothea_ab*"; "*trrust*"; "*hematopoiesis_76×108*" (76 hematopoiesis TFs × 108 immune / cross-tissue targets = 8,208 directed edges, used in Axis 4); and "*shared_36*" (the TRRUST-intersected activation/repression panel used for Axis 3 cross-dataset sensitivity). Axis 2 operates on all HVG pairs rather than a curated panel so that the coarse-versus-fine question is closed under a shared edge universe.

Edge scorers

Every axis pipeline accepts a pluggable "edge scorer" via a shared interface, so the same preprocessing and null library feeds every methodological choice. Five scorers are implemented:

- **Pearson correlation** on column-standardized expression. Sign of correlation is tracked separately from the magnitude score.
- **Mutual information** via the `mutual_info_regression` routine in scikit-learn (Pedregosa et al., 2011), which implements the Kraskov–Stögbauer–Grassberger k-nearest-neighbour estimator (Kraskov et al., 2004) ($k = 3$, seed fixed).
- **GRNBoost2** (Moerman et al., 2019) and **GENIE3** (Huynh-Thu et al., 2010): per-target gradient-boosted and random-forest regression importance respectively (500 estimators, sqrt feature subsampling).
- **scGPT attention** (Cui et al., 2024), extracted from the public *whole-human* checkpoint. For each cell we tokenise the panel genes, forward-pass through the twelve transformer layers, and average attention weights across all layers and heads; edge scores are the symmetrised cell-wise mean attention between each TF and target token.

Table 1. Datasets used across the five robustness axes. External lung = Travaglini et al. (2020) SmartSeq2 cohort (non-Tabula-Sapiens, non-10x technology), used to verify that conclusions survive outside the Tabula Sapiens pipeline. Delorey COVID lung = Delorey et al. 2021 COVID-19 lung atlas, providing a disease-state cohort and a large-scale validation (116,313 cells across 27 donors).

Dataset	Cells	Donors	Batches	Axes
Kidney	8,000–11,376	1	2	1, 2, 3, 5
Lung	15,000–20,000	4	7	2, 3, 4, 5
Immune	15,000–20,000	12–20	42	2, 3, 4, 5
Immune inv.	15,000	—	—	2, 3
External lung	9,000	—	—	2, 3
Delorey COVID lung	116,313	27	—	1, 2, 3

The scorer used for any figure or statistic is stated in the caption and §3.6 reports the pairwise agreement matrix. No model is trained or fine-tuned in this work: scGPT enters strictly as a frozen pretrained feature extractor, and the remaining scorers are unsupervised statistical estimators.

Metric definitions

For top-k index sets T_a^k, T_b^k from two scoring runs on a shared edge universe, the bounded-drift Resolution Sensitivity Score is $\mathrm{RSS}_k = w_o(1-\mathrm{overlap}_k) + w_j(1-\mathrm{Jaccard}_k) + w_d\,\mathrm{drift_norm}_k$, where $\mathrm{drift_norm}_k = \min(1, \mathrm{median}_{i \in T_a^k \cap T_b^k} |r_{a,i} - r_{b,i}|/k)$ maps median rank-shift to $[0,1]$. Unless noted, $k = 1{,}000$ and weights are $(0.4, 0.3, 0.3)$; the weight simplex is swept in §3.2. The remaining metrics are standard: Spearman rank stability, null-separation AUC, tail gap (top-5% quantile observed minus null), sign-flip rate, and mean absolute rank shift. A null family is *cleared* when the empirical p against it is ≤ 0.05 with ≥ 19 permutations.

Axis 1: Cell-count subsampling

Cell-count robustness was evaluated on all cohorts for which a sufficient cell count was available (kidney, lung, immune, immune-invariant, external lung, and Delorey COVID lung) at cell counts $\{200, 500, 1{,}000, 2{,}000, 3{,}000, 5{,}000, 8{,}000, 15{,}000\}$ (clipped to dataset maximum) with eight independent subsamples per size. An *anchor ablation* was added to address the concern that using a 3,000-cell reference is circular: each dataset's MVCC was computed independently against a 3,000-cell and an 8,000-cell anchor (plus a 15,000-cell anchor for Delorey, whose cell count permits it). Per-cell, per-anchor top-k Jaccard and overlap were computed at $k \in \{100, 500, 1{,}000, 5{,}000\}$, plus top-1%, top-5%, and top-10% proportions of the edge universe, and per-target top-5 and top-10 overlaps.

The Minimum Valid Cell Count (MVCC) is the smallest cell count at which the mean top-1,000 Jaccard exceeds 0.5 against the reference anchor. We report MVCC per (dataset, anchor, scorer) triple so that both anchor-dependence and scorer-dependence can be read directly from the results. Benjamini–Hochberg correction (Benjamini and Hochberg, 1995) was applied to the family of per-(policy, cell-count) p-values testing whether prior-constrained rankings differ from matched random-gene-set controls.

Axis 2: Cluster-resolution sensitivity

Across all six cohorts (Table 1), coarse labels were broad cell classes and fine labels were precomputed Leiden subclusters (Traag et al., 2019) enforced as strict refinements of the coarse partition. Edge scores combined global and per-group maximum absolute Pearson correlation, $S = 0.6R_{\mathrm{global}} + 0.4R_{\mathrm{group\,max}}$, over the top 600 HVGs; coarse-vs-fine agreement was the bounded-drift RSS of §2.2. A tiered calibration applied three null families (global cell-label shuffle, within-coarse constrained shuffle, degree-preserving bipartite rewire; 49 permutations each), downgrading the base RSS recommendation ("fine" ≤ 0.30, "hybrid" ≤ 0.45, else "coarse") one level per failed null family. RSS weights were also swept over the full 66-point simplex at step 0.1, an empirical null was constructed from 1,000 rank permutations per cohort, the HVG count was swept over $\{300, 600, 900, 1{,}200\}$, and the three normalization schemes from §2.1 were tested (Supplementary Fig. S6).

Axis 3: Rare-cell-type reliability

Cell types with ≥ 40 cells were tested under two pipelines: an adaptive-panel pipeline on three Tabula Sapiens tissues plus external lung (88 eligible cell types) and a shared-"primary"-panel pipeline on all five Tabula Sapiens cohorts (39 eligible cell types); Delorey COVID lung adds nine coarse compartments. Rarity was defined by within-dataset cell-count quartile. Eight bootstrap replicates at 80% subsampling were drawn per cell type and top-$k = 100$ edges scored, yielding rank stability (Spearman), top-k Jaccard, and null-separation AUC and tail gap under cell-permutation and gene-shuffle nulls. The pre-registered primary reliability gate was stability ≥ 0.75, Jaccard ≥ 0.35, AUC ≥ 0.65, positive tail gap, and cell count ≥ 80; these were fixed before a five-dimensional 5,880-point threshold-sensitivity grid was swept to test the 0% rare-passage headline.

Axis 4: Donor generalization

Donor generalization was assessed on immune (up to 13 donors at 200 cells/donor, or 8 at 300 cells/donor) and lung (4 donors, 800 cells/donor) using the hematopoiesis 76×108 panel (Supplementary Table S1; 8,208 directed edges before gene-universe restriction). A fixed-holdout design (two holdout donors held constant to eliminate train-size / holdout-size confounding) sampled training donor counts $\in \{2, 4, 6, 8, 10\}$ with 20 random splits per count. The cells-per-donor parameter was swept between 200 and 800. Transfer was quantified by Spearman ρ, Pearson r, top-k overlap and Jaccard at $k \in \{50, 100, 250, 500\}$, sign-flip rate, and mean absolute rank shift.

Axis 5: Integration-method sensitivity

Harmony (Korsunsky et al., 2019) was applied in PCA space (50 components) on immune, lung, and kidney using three batch keys (technical batch, donor, method); scVI (Gayoso et al., 2022) and Scanorama (Hie et al., 2019) were run as a cross-method comparison; mutual nearest-neighbour and graph-aligned alternatives such as MNN (Haghverdi et al., 2018) and

BBKNN (Polański et al., 2020) were not benchmarked here but are noted in the discussion as natural candidates for follow-up audits. Sensitivity was measured by rank Spearman, sign-flip rate, mean absolute rank shift, and top-k overlap relative to uncorrected data. External validation used precision@500 against DoRothEA AB / ABCD / ChIP-seq–backed subsets and TRRUST with paired swap binomial tests and BH-FDR correction; perturbation validation used Dixit-derived tables (Dixit et al., 2016). A $n_{\text{pcs}} \in \{30, 50, 80\}$ sweep probed the robustness–performance tradeoff.

Scorer-divergence and cross-axis analyses

To test transferability of fragility conclusions across scorers, all five scorers were run on the same 800 kidney cells under the hematopoiesis 76×108 panel (after restriction to kidney's gene universe: 75 of 76 TFs, 99 of 108 targets, 7,425 directed edges); pairwise agreement was reported via Spearman ρ, top-100 and top-500 Jaccard, and the bounded-drift RSS. A cross-axis fragility profile then summarised, per (axis, cohort), the severity of the robustness concern (pass / marginal / fail) under each axis' calibrated threshold.

Adversarial AI-assisted review protocol

Each axis passed through an iterative AI-assisted review loop that adversarially challenged the initial analysis under a fixed rubric (confounded nulls, overfitted hyperparameters, leakage, composite-metric uninterpretability, threshold sensitivity); corrective experiments were designed by the agent, executed, and verified before any claim was updated. The base model, prompt templates, round-by-round logs, and the 27-issue full-census audit under the frozen rubric are in Supplementary Methods S2.

Results

Axis 1: The minimum valid cell count is strongly scorer-dependent and anchor-invariant

Under scGPT attention, even moderate cell-count reductions reshape top-ranked edges: at 200 cells on kidney, top-1,000 Jaccard against the 3,000-cell anchor is 0.001–0.029, and at 1,000 cells it remains below 15%, so the scGPT-attention MVCC exceeds 3,000 cells.

The threshold is scorer-specific. Under Pearson correlation on 600 HVGs (Figure 1A), the MVCC (smallest cell count at which mean top-1,000 Jaccard exceeds 0.5) is **200 cells for kidney, lung, immune-invariant, and external lung** and **500 cells for primary immune and Delorey COVID lung**. MVCC is *anchor-invariant* (Figure 1B; Supplementary Fig. S1): identical values emerge against 3,000-, 8,000-, and (Delorey only) 15,000-cell anchors, ruling out anchor-threshold circularity. Kidney Jaccard climbs 0.515 (200 cells) $\rightarrow$ 0.747 (1,000) $\rightarrow$ 0.839 (3,000) at the 3,000-cell anchor; other cohorts follow similar trajectories (Supplementary Fig. S5).

Constraining scGPT-attention scoring to DoRothEA TF–target pairs stabilises kidney rankings dramatically: the two TF-guided policies hold Jaccard at 0.73–0.76 (200 cells) and 0.86–0.87 (1,000 cells) versus 0.002–0.008 for matched random gene-set controls (BH-corrected $q \approx 0.017$; Figure 1B), and recover 5.5–6.6% of top-1,000 edges with independent DoRothEA support (versus 0.2–0.3% for controls). Prior-constrained policies therefore keep rankings meaningful far below the scGPT-specific MVCC. The guidance is: *plan for* $\geq$ 3,000 *cells under scGPT or use prior constraints; 200–500 cells suffice under Pearson, MI, and tree scorers on the cohorts tested.* §3.6 examines scorer-dependence directly.

Axis 2: Dual-null calibration pushes every cohort to coarse or hybrid clustering; the recommendation is robust to weight choice, gene dimension, normalization, and the addition of a third null family

Does coarse-versus-fine clustering change the resulting GRN? Observed bounded-drift RSS at $k = 1{,}000$ is 0.218 (immune-invariant) to 0.402 (lung) (Figure 2A). A 1,000-permutation empirical null gives mean ≈ 0.79 and 5th percentile ≈ 0.73, placing every cohort 8.8–13.1 standard deviations below the null mean ($p = 0.001$ floor; Supplementary Fig. S3): every cohort is therefore significantly more stable than a random coarse-vs-fine pairing.

Dual-null calibration refines the picture. Only kidney, lung, and immune-invariant clear the global cell-label shuffle ($p = 0.02$); *no* cohort clears the within-coarse constrained shuffle ($p \geq 0.14$ across all five Tabula Sapiens cohorts). Requiring both nulls for any "fine" or "hybrid" recommendation downgrades four Tabula Sapiens cohorts to **coarse** and immune-invariant to **hybrid** (Figure 2B); the apparent fine-stability of immune-invariant and external lung evaporates once the within-coarse null is applied. Delorey COVID (116,313 cells) has the lowest raw RSS of any cohort (0.173) but fails the within-coarse null at $p = 0.69$ and therefore also downgrades to **hybrid**.

The dual-null recommendation is invariant to weight choice (66-point simplex sweep at step 0.1, Supplementary Fig. S2), HVG count ($n \in \{300, 600, 900, 1{,}200\}$; immune-invariant flips coarse$\leftrightarrow$hybrid but never to fine), and normalization scheme (depth-log1p / size-factor-log1p / Pearson residuals on kidney all give coarse; Supplementary Fig. S6). A triple-null calibration adds a permissive degree-preserving bipartite rewire as a third null: every cohort clears it at the permutation floor ($p = 0.02$) while still failing within-coarse at $p \geq 0.22$, so the within-coarse failure reflects genuine over-partitioning rather than null over-stringency (Supplementary Fig. S7).

Within-class heterogeneity remains considerable even after coarsening: T cells in kidney (RSS 0.466) and immune (0.314), erythroid lineage in immune (0.535) and immune-invariant (0.522), and B-lineage lymphocytes in kidney (0.597) are markedly more fine-sensitive than their surrounding compartments (full per-compartment breakdown in the reproduction package).

Axis 3: Rare cell types exhibit a reliability gap that is robust to 5,880 threshold combinations

Are edges inferred from rare cell types as trustworthy as those from abundant types? Across 88 cell types in the four adaptive-panel datasets (three Tabula Sapiens tissues plus external lung), rare types match abundant types on overall rank ordering (Spearman 0.884 rare vs 0.851 abundant, $p = 0.835$) but are significantly worse on the decision-relevant top-k overlap measures: Jaccard 0.625 vs 0.731 ($p = 4.1 \times 10^{-3}$), null-separation AUC 0.520 vs 0.620 ($p = 5.3 \times 10^{-8}$), tail gap -0.062 vs $+0.082$ ($p = 1.5 \times 10^{-7}$). Under the pre-registered reliability gate (stability ≥ 0.75, Jaccard ≥ 0.35, primary AUC ≥ 0.65, positive tail gap, cell count ≥ 80), **0 of 22 rare cell types pass versus 4 of 22 abundant types** (Figure 3A–B); under a dual-null formulation (minimum AUC across cell-permutation

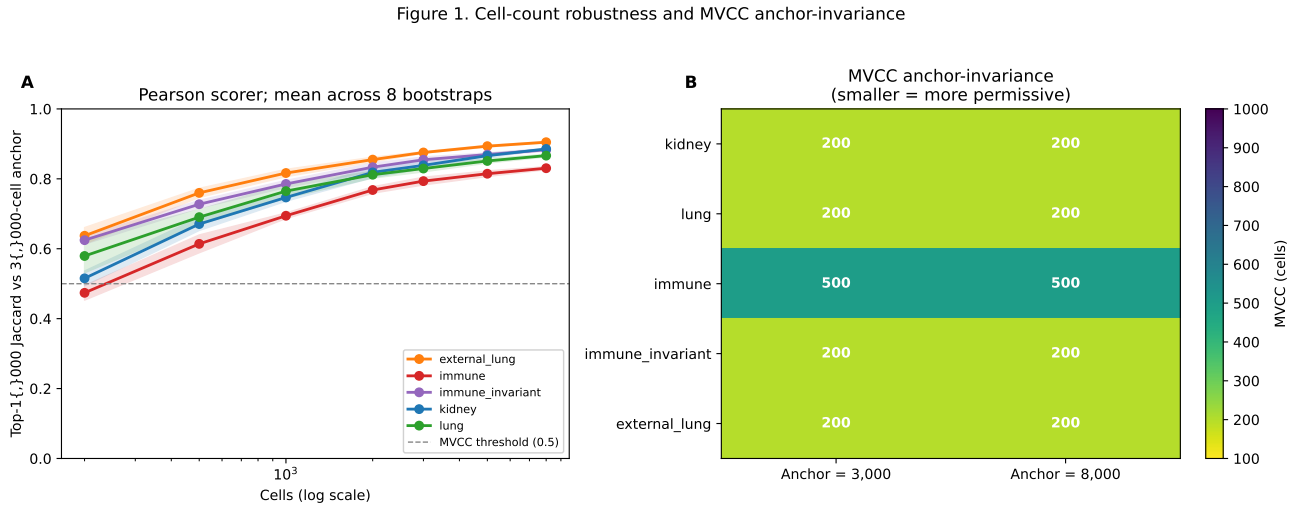

Fig. 1. Cell-count subsampling robustness (Pearson scorer). (A) Top-1,000 reference Jaccard as a function of cell count for each cohort (Pearson correlation over 600 HVGs), computed against a 3,000-cell anchor. Shaded bands: standard deviation across eight independent subsamples per size. Dashed horizontal line: the MVCC Jaccard threshold (0.5). (B) MVCC (smallest cell count clearing the 0.5 Jaccard threshold) reported against two reference anchors per cohort, demonstrating anchor-invariance. Full per-scorer Jaccard-vs-cells curves are in Supplementary Fig. S1.

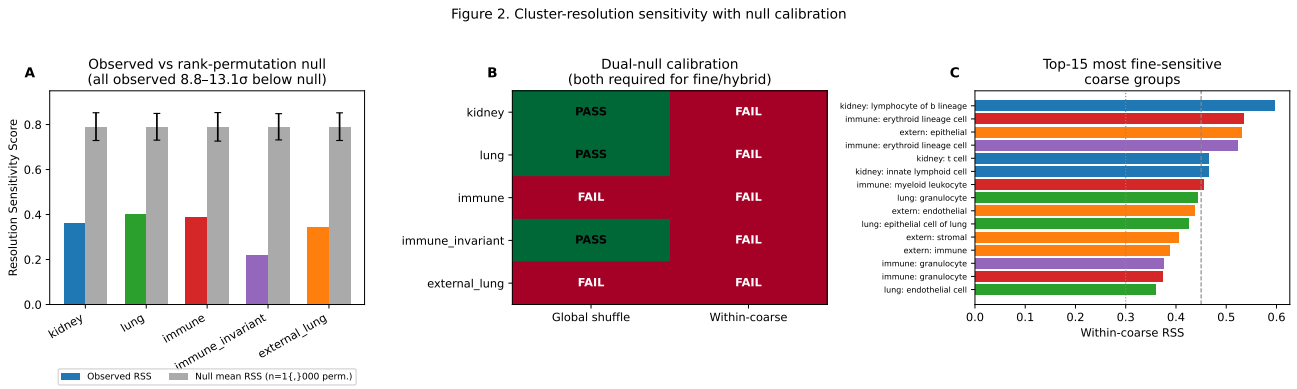

Fig. 2. Cluster-resolution sensitivity and dual-null calibration. (A) Observed bounded-drift Resolution Sensitivity Score (coloured bars) versus the rank-permutation null mean and 5th–95th percentile (grey error bars) for each Tabula Sapiens cohort; every observed value sits 8.8–13.1 standard deviations below the null. (B) Dual-null pass/fail heatmap: only kidney, lung, and immune-invariant clear the global cell-label shuffle; no cohort clears the within-coarse constrained shuffle. (C) Top-15 most fine-sensitive coarse groups by within-coarse RSS; dotted lines mark the base-RSS "fine" (0.30) and "hybrid" (0.45) thresholds. Colour encodes the cohort of origin.

and gene-shuffle nulls), the gap attenuates but stays significant (AUC gap 0.018, $p = 6.8 \times 10^{-4}$).

A 5,880-point threshold grid (5-D sweep over stability, Jaccard, AUC, tail gap, and cell count) confirms the headline is not threshold-driven: the abundant->-rare gap direction holds at *100%* of grid points, the grid-mean gap is +0.344 (wider than +0.182 at primary thresholds), and rare pass rate is 0 at two-thirds of grid points (Supplementary Fig. S4). Switching to a shared 36-edge TRRUST-intersected panel preserves the gap direction on immune and kidney (Spearman 0.93 and 0.87 for null-AUC ordering) but not lung (0.09); the shared "primary" panel (DoRothEA A/B ∩ TRRUST) on all five Tabula Sapiens cohorts also preserves it (0% rare / 5.9% intermediate / 9.1% abundant; Supplementary Table S3). Minimum reliable cell-size estimates under the adaptive-panel pipeline range from 287 (external lung, extrapolated) to 16,486 (lung, empirical), so some tissues require very large cell-type samples before mechanistic edges clear the reliability gate (Figure 3C).

Disease-cohort validation.

Delorey COVID lung (116,313 cells, 27 donors) presents a qualitatively different failure mode: across nine coarse compartments, rank stability is high (Spearman 0.85–0.89) and Jaccard reasonable (0.53–0.63), but null-separation AUC is uniformly weak (0.50–0.64), so **no cell type passes the gate in any rarity class** (0 of 3 each). Disease-driven shared expression programmes (COVID-induced myeloid / fibroblast remodelling) plausibly compress the observed-vs-null gap; the gate was calibrated on Tabula Sapiens and may need disease-specific recalibration for heavily perturbed tissues.

Axis 4: Donor generalization improves monotonically; the required donor count depends on cells-per-donor balancing

Under the fixed-holdout protocol (two holdout donors held constant), immune-tissue transfer improves monotonically with

Figure 3. Rare-cell-type reliability and threshold sensitivity

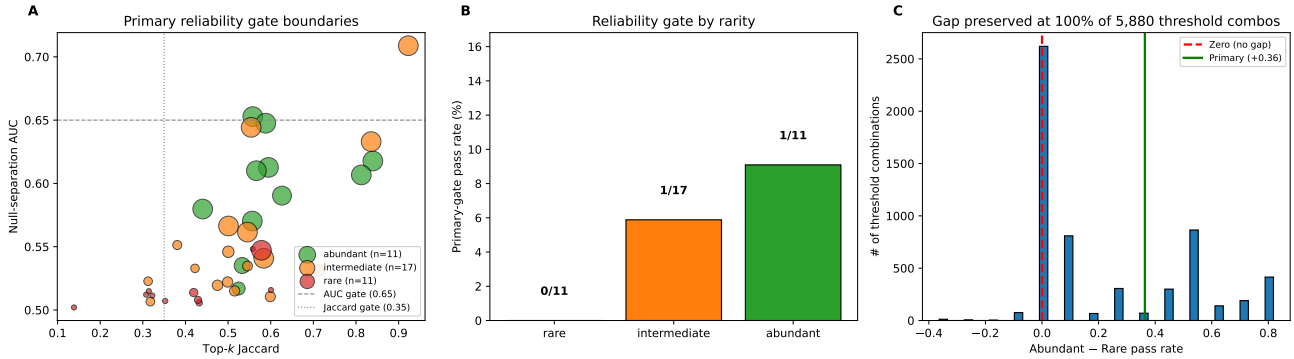

Fig. 3. Rare-cell-type reliability and threshold sensitivity. (A) Scatter of null-separation AUC versus top-k Jaccard for every eligible cell type, coloured by within-dataset rarity group (rare / intermediate / abundant). Marker size scales with cell count. Dashed grey lines mark the primary-gate thresholds (AUC $\geq$ 0.65, Jaccard $\geq$ 0.35); points in the upper-right quadrant are candidates for passing. (B) Primary-gate pass rates by rarity on the shared "primary" panel (fragility package output); numerators/denominators shown above each bar. (C) Histogram of the abundant-minus-rare pass-rate gap across all 5,880 threshold combinations on the five-dimensional grid; the primary-threshold gap is marked in green and zero (no gap) in red. The gap direction is preserved at 100% of grid points.

training-donor count. At 300 cells/donor in the 20k-cell immune subset, Spearman ρ rises 0.568 $\rightarrow$ 0.683 from 2 to 10 training donors and top-100 overlap rises 0.445 $\rightarrow$ 0.540 (Figure 4A). Expanding to the top-20 donors (long-tail inclusion) lowers all transfer metrics (ρ = 0.627 at 16 donors vs 0.683 at 10 in the top-12 panel), showing that high-quality-donor restrictions paint an optimistic picture. The 90%-of-best criterion recommends **6 training donors**; the 95%-of-best criterion recommends **8**. Under looser balancing (200 cells/donor, 13 eligible donors), the curve shifts and 90%-of-best is met at 2 donors — the requirement is therefore *cells-per-donor dependent*, and the cells-per-donor policy must be pre-registered.

Composition matching (3 shared cell classes, 62 cells/donor) lowers transfer by 16.3% on Spearman (0.707 $\rightarrow$ 0.591) and 10.7% on top-100 overlap (0.610 $\rightarrow$ 0.545; Figure 4B): a substantial fraction of apparent donor generalisation reflects shared cell-type composition. Even under this strictest protocol the above-null signal remains large (lift@100 of 37.8–44.3$\times$), indicating real cross-donor regulatory signal beyond composition.

Axis 5: Integration method choice reorders rankings substantially and does not improve perturbation-grounded validity

Harmony integration produces moderate-to-large rank perturbations on all three tissues. On the full HVG-pair universe (unconstrained), Spearman corrected-vs-uncorrected ranges 0.757–0.823 with sign-flip rates 8.1–12.7%. Panel-restricted (hematopoiesis 76 $\times$ 108): top-500 overlap 65–71%, top-1,000 overlap 72–75%, i.e. 29–35% top-500 displacement (Figure 5A) — roughly a third of prioritised edges change depending on whether batch correction is applied. Yet rankings stay highly non-random: bounded-drift RSS between baseline and Harmony-integrated lists is 0.26–0.30 across the eight (tissue $\times$ batch-key) combinations vs null mean $\approx$ 0.72 ($|z| \geq 37$; empirical p at floor).

Integration method matters.

A cross-method comparison (Harmony vs scVI vs Scanorama, same cells, same panel, same random seed) reveals that method choice is not neutral (Table 2). Three regimes emerge:

- **Harmony** preserves rankings reasonably well across all three tissues (Spearman 0.63–0.87, top-1,000 overlap 0.65–0.87, sign-flip 6–17%).
- **scVI** sits in the middle: Spearman 0.36–0.57, top-1,000 overlap 0.48–0.74, sign-flip 18–30%. Roughly a quarter of edges change sign compared to the uncorrected baseline.
- **Scanorama** produces near-random rankings on kidney and lung (Spearman $\approx$ 0, top-1,000 overlap 0.11–0.14, sign-flip 44–47%; RSS at the rank-permutation null mean of $\approx$ 0.72) and weakly correlated rankings on immune (Spearman 0.10, top-1,000 overlap 0.62, sign-flip 26%).

In other words, switching integration method on the same data can cause anywhere from 6% to 47% of the top-500 edges to change sign, and can move the top-1,000 ranking to statistical indistinguishability from a random rewire of the baseline. Any GRN claim downstream of integration therefore depends materially on integration-method choice; we recommend reporting rank-stability metrics against the uncorrected baseline for every integration tool used.

Do integrated rankings better recover reference edges?

Raw precision@500 shows nominal Harmony gains on DoRothEA and TRRUST in immune and kidney, but paired swap tests with BH-FDR correction restrict robust improvements to TRRUST on immune and kidney (Δ precision +0.026 to +0.028, FDR $<$ 0.05; Figure 5B); DoRothEA and ChIP-seq–backed gains do not survive correction. Perturbation validation against Perturb-seq (Dixit et al., 2016; Replogle et al., 2022) returns small, negative top-500 swap deltas (-0.002 to -0.004, FDR = 1.0): edges *gained* by integration are no more likely to be experimentally validated than edges *lost*. Ranking perturbation does not translate into biological gain even for the best-behaved integrator. A $n_{\text{pcs}} \in \{30, 50, 80\}$ sweep reveals a tradeoff: top-500 overlap with baseline rises monotonically with PCs (immune 0.590 $\rightarrow$ 0.736, more

Figure 4. Donor generalisation under fixed-holdout

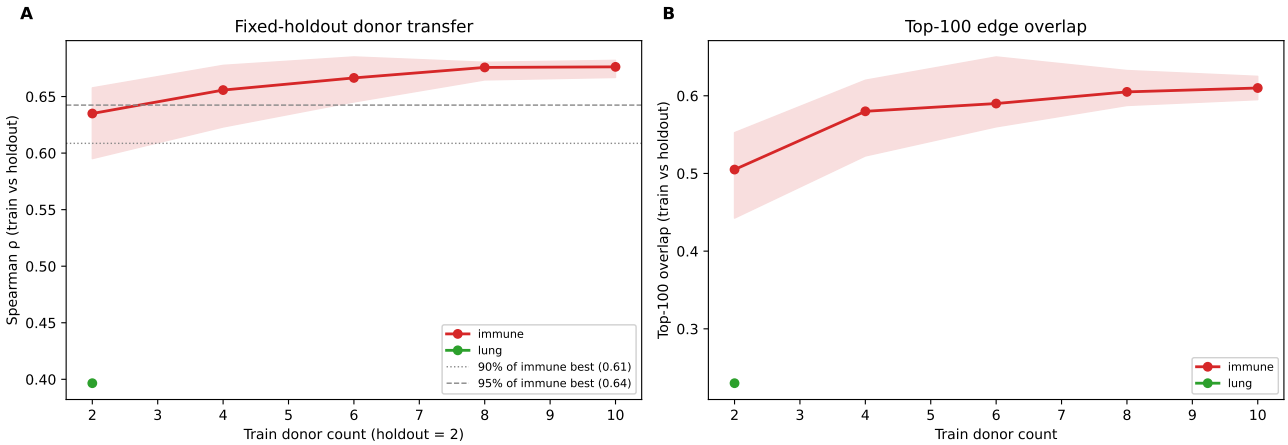

Fig. 4. Donor generalization under fixed-holdout and composition control. (A) Immune transfer curves under fixed-holdout (holdout= 2) for top-12 and top-20 donor panels. (B) Composition-matched versus unmatched transfer in the top-8 panel. Error bands show interquartile range over 30–40 sampled splits.

Table 2. Cross-method integration perturbation of edge rankings. All rows: 1,200 cells, hematopoiesis 76 × 108 panel, Pearson scorer, same random seed. Top-1,000 overlap below ∼ 0.3 with RSS at the rank-permutation null mean ≈ 0.72 indicates near-random rankings.

Dataset	Method	Spearman	Overlap@500	Overlap@1k	Sign-flip	RSS@500
Kidney	Harmony	0.78	0.62	0.69	15%	0.38
Kidney	scVI	0.57	0.42	0.51	24%	0.51
Kidney	Scanorama	-0.04	0.08	0.11	47%	0.76
Lung	Harmony	0.63	0.59	0.65	17%	0.42
Lung	scVI	0.36	0.42	0.48	30%	0.54
Lung	Scanorama	-0.02	0.08	0.14	44%	0.77
Immune	Harmony	0.87	0.81	0.87	6%	0.20
Immune	scVI	0.54	0.63	0.74	18%	0.37
Immune	Scanorama	0.10	0.32	0.62	26%	0.60

conservative integration), but DoRothEA precision peaks at intermediate PCs (Figure 5C), so no single setting maximises both stability and reference agreement.

Scorer divergence: within-scorer fragility does not transfer across scorer families

Scoring the hematopoiesis 76 × 108 panel on a shared 800-cell kidney sample with all five scorers reveals the pairwise agreement landscape (Table 3, Figure 6). The two tree-ensemble scorers (GRNBoost2 and GENIE3) are near-identical (Spearman $\rho = 0.93$, top-500 Jaccard = 0.54); Pearson is moderately correlated with the tree scorers ($\rho \in [0.52, 0.56]$); mutual information is weakly correlated with Pearson ($\rho = 0.20$) and unrelated to the tree scorers ($\rho \leq 0.21$). Most importantly, **scGPT attention is uncorrelated with every classical scorer**: Spearman ρ ranges -0.05 (vs Pearson) to $+0.20$ (vs MI), top-100 Jaccard 0.02–0.06, and the RSS composite 0.69–0.76 sits at or above the rank-permutation null mean (≈ 0.72). The Axes 1–5 within-scorer fragility analysis is therefore orthogonal to across-scorer agreement: choosing a stable scorer does not guarantee another stable scorer will agree with it, and an analyst switching scorers must re-audit.

Cross-axis analysis

The five axes are not independent (Table 4, Figure 7). Three interactions matter for practice. First, Axis 1 and Axis 3 are mechanistically linked under scGPT attention: rare types (< 500 cells) operate below the scGPT MVCC of ∼3,000, partly explaining their 0% reliability-gate passage; under Pearson and tree scorers (MVCC 200–500) the link is weaker, so the rare-vs-abundant gap is also driven by sparsity and lower null-separation in small subsamples. Second, the dual-null coarse recommendation from Axis 2 indirectly aids Axis 4, since coarse clustering pools cell-type variation that would otherwise require per-donor composition matching. Third, Axis 5 operates upstream of every other axis by altering the correlation structure on which scores are computed, so integration choice is a latent confounder for cell-count, resolution, and rare-type calibrations. No dataset passes all five axes under their calibrated criteria in any single scorer family, no single corrective measure resolves all fragilities, and within-scorer fragility profiles do not transfer across scorer families: robustness must be demonstrated, not assumed, along each axis *and* each scorer choice.

Discussion

This audit demonstrates that GRN edge rankings from single-cell data are fragile along multiple axes simultaneously, and

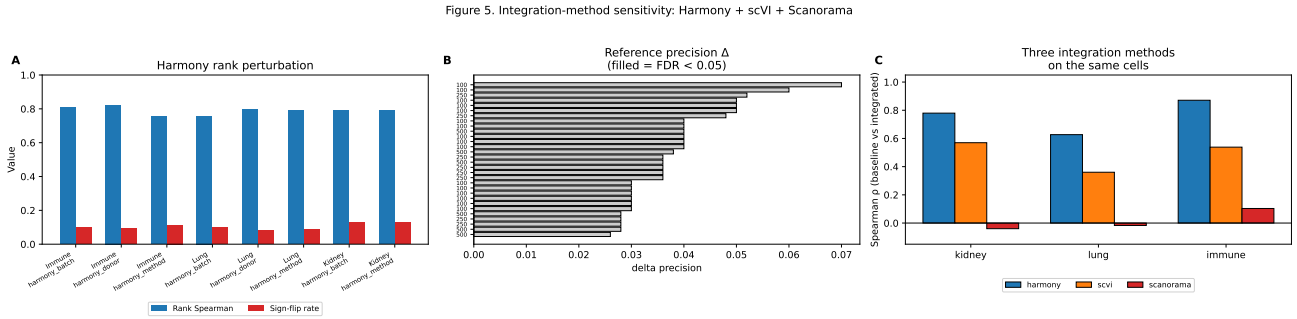

Fig. 5. Integration-method sensitivity. (A) Rank Spearman and sign-flip rate for Harmony variants (tissue × batch key combinations) against uncorrected baseline. (B) Reference precision@500 deltas for Harmony-integrated vs uncorrected rankings, with paired swap-based binomial significance tests (filled bars: FDR < 0.05 under Benjamini–Hochberg correction; unfilled: not significant). (C) Cross-method comparison of three integration tools: Harmony (blue), scVI (orange), and Scanorama (red), on 1,200 cells from each tissue (Pearson scorer, hematopoiesis 76×108 panel). Dashed horizontal line at $\rho = 0$ indicates rank-permutation null.

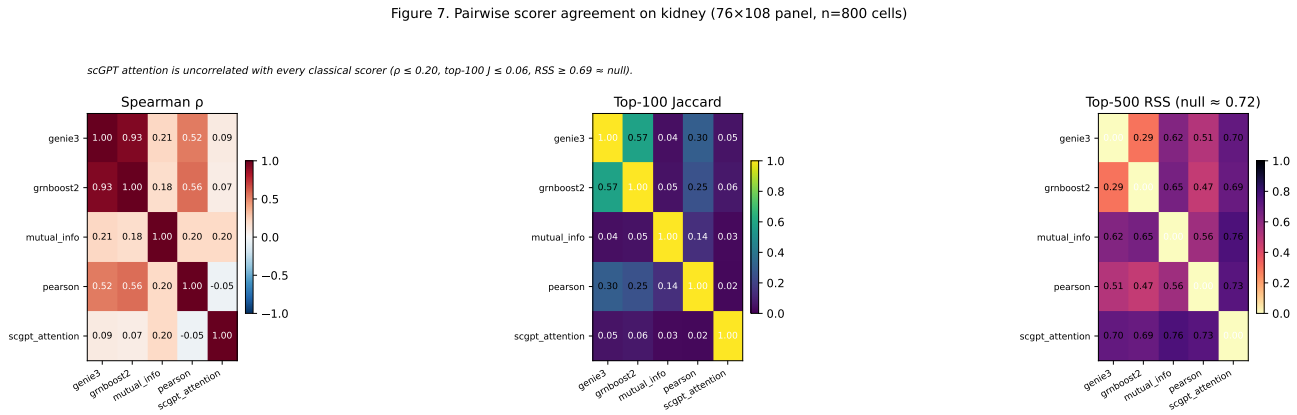

Fig. 6. Pairwise scorer agreement on kidney (hematopoiesis 76×108 panel, $n = 800$ cells). Left: Spearman ρ. Centre: top-100 Jaccard. Right: top-500 RSS composite (null mean ≈ 0.72, shown in darker red). scGPT attention is uncorrelated with every classical scorer (last row/column); GRNBoost2 and GENIE3 are nearly identical.

that apparent robustness along one axis (e.g., coarse rank stability for rare types) can mask failure along another (e.g., null separation). The adversarial-review methodology that we applied to each axis is itself a contribution: by requiring corrective experiments before claims are updated, it systematically closes the gap between initial findings and defensible conclusions.

Practical implications.
Table 5 provides concrete guidance: no single parameter choice can guarantee robust GRN inference; robustness must be demonstrated along each relevant axis for the specific tissue, cell count, and analytical choices at hand.

Comparison to existing benchmarks.
Published GRN benchmarks (Pratapa et al., 2020; Chen and Mar, 2018) have focused primarily on accuracy relative to gold-standard references, treating preprocessing as fixed. Our audit complements these by asking whether rankings are *stable* across realistic analytical perturbations, a necessary condition for practical utility. The two perspectives are not redundant: a method can be accurate under one configuration and fragile under perturbation, or robustly mediocre across all configurations.

Limitations.
None of the five scorers directly estimate causal regulatory effects; correlation on normalized counts avoids zero-inflation artifacts (Svensson, 2020) but remains susceptible to confounding by shared upstream regulators (Kharchenko et al., 2014), tree-ensemble scorers introduce their own biases, and scGPT attention reflects pretraining patterns on the CellxGene Census rather than causal edges. The audit covers four Tabula Sapiens tissues plus one external-lab SmartSeq2 lung cohort and one COVID-19 lung disease cohort; generalisation to other disease tissues, developmental atlases, and cross-species comparisons remains untested. The within-coarse null is demanding and other scoring families might clear it; the single-cell-DE literature documents how easily under-calibrated nulls inflate apparent discoveries (Squair et al., 2021), supporting the within-coarse stringency we adopt. Perturbation-backed validation remains data-sparse, limiting swap-test power. Three integration methods are compared side-by-side (Harmony (Korsunsky et al., 2019), scVI (Gayoso et al., 2022), Scanorama (Hie et al., 2019)); broader integration benchmarks (Luecken et al., 2022; Tran et al., 2020) and recent multimodal integration frameworks (Hao et al., 2024) suggest extending the cross-method comparison further is a natural next step.

Table 3. Pairwise scorer-agreement matrix on kidney (76 TFs × 99 targets panel, $n = 800$ cells). All top-500 RSS values should be compared against the rank-permutation null mean of ≈ 0.72: values far below null indicate concordant rankings, values near null indicate ranking independence.

Scorer A	Scorer B	Spearman	Top-100 J	Top-500 J	Top-500 RSS
Pearson	Mutual info	0.20	0.14	0.21	0.56
Pearson	GRNBoost2	0.56	0.25	0.32	0.47
Pearson	GENIE3	0.52	0.30	0.26	0.51
Pearson	scGPT attention	−0.05	0.02	0.04	**0.73**
Mutual info	GRNBoost2	0.18	0.05	0.14	0.65
Mutual info	GENIE3	0.21	0.04	0.16	0.62
Mutual info	scGPT attention	0.20	0.03	0.06	**0.76**
GRNBoost2	GENIE3	0.93	0.58	0.54	0.29
GRNBoost2	scGPT attention	0.07	0.06	0.06	**0.69**
GENIE3	scGPT attention	0.09	0.05	0.05	**0.70**

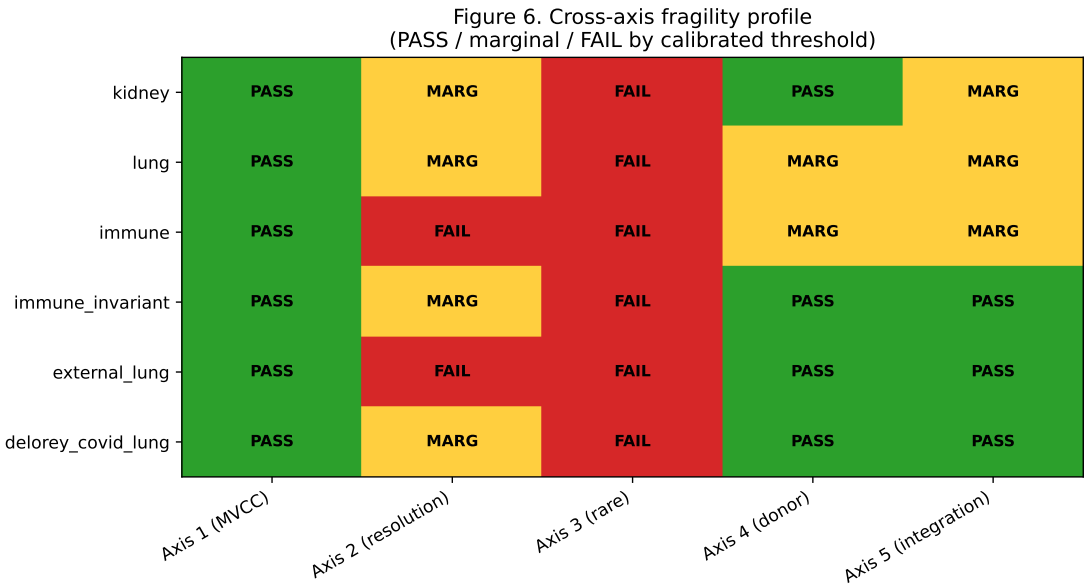

Fig. 7. Cross-axis fragility summary. Heatmap showing pass (green), marginal (yellow), and fail (red) status for each axis–dataset combination under calibrated criteria.

Conclusion

This systematic multi-scorer, multi-axis robustness audit of single-cell GRN inference — five scorers, six cohorts, five axes (cell count, clustering resolution, rare-cell reliability, donor generalisation, integration sensitivity) — finds fragility to be pervasive, axis-specific, interactive, and scorer-specific. No dataset passes all five axes under calibrated criteria in any scorer family; the within-coarse resolution null fails universally under dual- and triple-null calibrations; the rare-vs-abundant reliability gap holds across 5,880 threshold combinations; and scGPT attention rankings are statistically uncorrelated with classical-scorer rankings on the same cells. The accompanying `fragility` package reproduces every CSV and figure from raw h5ad via a single `make audit` command and is extensible to new scorers and datasets through a single interface; we encourage practitioners to conduct analogous audits before publishing GRN claims.

Author Contributions

Ihor Kendiukhov: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration.

Acknowledgements

The author thanks the Tabula Sapiens Consortium and the authors of the Travaglini et al. 2020 lung atlas and the Delorey et al. 2021 COVID-19 lung atlas for making their single-cell data publicly available, and the authors of scGPT and the DoRothEA / TRRUST databases for releasing their tools and references under open licences.

Funding

This work was not supported by any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. No external funding was received.

Table 4. Per-axis robustness summary: calibrated criterion and practical recommendation.

Axis	Calibrated criterion	Recommendation
Cell count	MVCC (top-1k Jaccard > 0.5); scorer-dependent: 200 cells (Pearson / tree, 4 of 6 cohorts), 500 cells (Pearson, primary immune and Delorey COVID lung), >3,000 cells (scGPT attention, anchor-invariant)	Use the MVCC for your chosen scorer family; apply DoRothEA-constrained policies to rescue small-sample scGPT runs
Resolution	Dual-null pass: both global cell-label shuffle and within-coarse constrained shuffle must clear at empirical $p \leq 0.05$	Default to coarse; report within-class RSS heterogeneity
Rare types	Reliability gate: stability ≥ 0.75, top-k Jaccard ≥ 0.35, null AUC ≥ 0.65, positive tail gap, cell count ≥ 80	Treat rare-cell-type edges as hypothesis-generating only
Donor	Fixed-holdout Spearman $\geq 90\%$ of best (300 cells/donor)	Collect ≥ 6 donors (8 for conservative generalisation)
Integration	Paired-swap Δ precision against curated reference, BH-FDR < 0.05	Report multi-reference validation; avoid universal claims of biological gain

Table 5. Practical guidelines for robust GRN inference from single-cell data.

Decision Point	Guideline
Sample size	Target $\geq$ 3,000 cells under scGPT attention or use informative prior constraints (DoRothEA / TRRUST) which stabilise scGPT rankings at 200+ cells. Under Pearson, mutual-information, and tree-based scorers, 200–500 cells are often sufficient on the cohorts tested; verify per-scorer MVCC before committing.
Clustering	Default to coarse clustering unless a dual-null calibration (global + within-coarse constrained shuffle) confirms that fine resolution adds separable signal; in our six cohorts, no coarse-vs-fine configuration cleared both nulls, so every recommendation collapses to coarse or hybrid. Report within-class RSS for compartment-level uncertainty.
Rare cell types	Report null-separation AUC alongside rank stability for every cell type; a high Spearman stability can coexist with null-level AUC (Delorey COVID cohort, §3.3). Flag rare types ($<$ 250 cells) as hypothesis-generating. Provide minimum-reliable-size estimates with uncertainty status.
Donor coverage	Collect ≥ 6 donors under strict cells-per-donor balancing (300 cells/donor, 8 donors for conservative generalisation); under looser balancing (200 cells/donor), the requirement drops to 2–4 donors. State the cells-per-donor policy explicitly. Use fixed-holdout evaluation, not complementary splits, and report composition-controlled transfer alongside raw transfer.
Integration	Run at least two integration methods on the same cells, not only the tool that happens to be most popular in the pipeline: we observe Spearman ranging from 0.06 (Scanorama on kidney) to 0.87 (Harmony on immune) on identical inputs. Report at least paired swap significance for reference gains, plus one perturbation-backed metric when available, and include a hyperparameter-sensitivity sweep. Avoid universal claims of biological improvement.
Scorer choice	Auditing one scorer does not license conclusions about another: in our kidney experiment, scGPT-attention rankings overlap with Pearson rankings in fewer than 5% of top-100 edges. Re-audit every axis under the scorer used to generate the published rankings.
Cross-axis audit	Report a fragility profile (Table 4) covering all axes relevant to the study design. Acknowledge which axes were not testable and why.

Conflict of Interest

None declared.

Data and Code Availability

The unified `fragility` package (source code, analysis pipelines, null library, metric definitions, panel registry, scorer implementations, and figure-regeneration scripts) is publicly available at `https://github.com/Biodyn-AI/grn-robustness-audit` and a versioned archive of the code accompanying this submission is deposited on Zenodo at `https://doi.org/10.5281/zenodo.20009065`. A single `make audit` command reproduces every CSV and figure in this paper from raw h5ad input.

Raw single-cell data are from the following public sources: Tabula Sapiens Consortium (`https://tabula-sapiens-portal.ds.czbiohub.org/`); Travaglini et al. 2020 SmartSeq2 lung atlas (CellxGene); and Delorey et al. 2021 COVID-19 lung atlas (CellxGene dataset `75c059c8-8fb7-4e6e-a618-a3e01ac42060`). The scGPT *whole-human* checkpoint is from Cui et al. (2024); the DoRothEA and TRRUST panels are from Garcia-Alonso et al. (2019) and Han et al. (2018) respectively.

Supplementary Information

Supplementary Methods S1 contains the panel registry (gene lists for every named TF–target panel used in the paper) and extended preprocessing documentation. Supplementary Methods S2 describes the adversarial AI-assisted review protocol, including base model, prompt rubric, round-by-round logs, and the full-census 27-issue classification under the pre-registered rubric. Supplementary Figures S1–S7 cover, respectively: the full MVCC anchor ablation; the RSS weight-simplex sweep; the rank-permutation empirical null distribution; the 5,880-point reliability-gate threshold-sensitivity heatmap; the top-k scan over $k \in \{100, 500, 1{,}000, 5{,}000, 1\%, 5\%, 10\%\}$; the normalization ablation (depth / size-factor / Pearson residuals); and the

triple-null calibration. Supplementary Table S3 reports Axis-3 per-cell-type reliability metrics under the shared primary panel for all Tabula Sapiens cohorts. All supplementary material is available at *Bioinformatics* online and in the GitHub repository.

Abbreviations

AUC: Area Under the Curve; BH: Benjamini–Hochberg; FDR: False Discovery Rate; GRN: Gene Regulatory Network; MVCC: Minimum Valid Cell Count; PCA: Principal Component Analysis; RSS: Resolution Sensitivity Score; scRNA-seq: Single-Cell RNA Sequencing; TF: Transcription Factor.

References

Aibar, S., González-Blas, C. B., Moerman, T., et al. (2017). SCENIC: single-cell regulatory network inference and clustering. *Nature Methods*, 14(11):1083–1086.

Benjamini, Y. and Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300.

Chen, S. and Mar, J. C. (2018). Evaluating methods of inferring gene regulatory networks highlights their lack of performance for single cell gene expression data. *BMC Bioinformatics*, 19(1):232.

Cui, H., Wang, C., Maan, H., et al. (2024). scGPT: toward building a foundation model for single-cell multi-omics using generative AI. *Nature Methods*, 21(8):1470–1480.

Delorey, T. M., Ziegler, C. G. K., Heimberg, G., et al. (2021). COVID-19 tissue atlases reveal SARS-CoV-2 pathology and cellular targets. *Nature*, 595(7865):107–113.

Dixit, A., Parnas, O., Li, B., et al. (2016). Perturb-Seq: dissecting molecular circuits with scalable single-cell RNA profiling of pooled genetic screens. *Cell*, 167(7):1853–1866.e17.

Garcia-Alonso, L., Holland, C. H., Ibrahim, M. M., Turei, D., and Saez-Rodriguez, J. (2019). Benchmark and integration of resources for the estimation of human transcription factor activities. *Genome Research*, 29(8):1363–1375.

Gayoso, A., Lopez, R., Xing, G., et al. (2022). A Python library for probabilistic analysis of single-cell omics data. *Nature Biotechnology*, 40(2):163–166.

Haghverdi, L., Lun, A. T. L., Morgan, M. D., and Marioni, J. C. (2018). Batch effects in single-cell RNA-sequencing data are corrected by matching mutual nearest neighbors. *Nature Biotechnology*, 36(5):421–427.

Han, H., Cho, J.-W., Lee, S., et al. (2018). TRRUST v2: an expanded reference database of human and mouse transcriptional regulatory interactions. *Nucleic Acids Research*, 46(D1):D380–D386.

Hao, Y., Stuart, T., Kowalski, M. H., et al. (2024). Dictionary learning for integrative, multimodal and scalable single-cell analysis. *Nature Biotechnology*, 42(2):293–304.

Heumos, L., Schaar, A. C., Lance, C., et al. (2023). Best practices for single-cell analysis across modalities. *Nature Reviews Genetics*, 24(8):550–572.

Hie, B., Bryson, B., and Berger, B. (2019). Efficient integration of heterogeneous single-cell transcriptomes using Scanorama. *Nature Biotechnology*, 37(6):685–691.

Huynh-Thu, V. A., Irrthum, A., Wehenkel, L., and Geurts, P. (2010). Inferring regulatory networks from expression data using tree-based methods. *PLoS ONE*, 5(9):e12776.

Kharchenko, P. V., Silberstein, L., and Scadden, D. T. (2014). Bayesian approach to single-cell differential expression analysis. *Nature Methods*, 11(7):740–742.

Korsunsky, I., Millard, N., Fan, J., et al. (2019). Fast, sensitive and accurate integration of single-cell data with Harmony. *Nature Methods*, 16(12):1289–1296.

Kraskov, A., Stögbauer, H., and Grassberger, P. (2004). Estimating mutual information. *Physical Review E*, 69(6):066138.

Lause, J., Berens, P., and Kobak, D. (2021). Analytic Pearson residuals for normalization of single-cell RNA-seq UMI data. *Genome Biology*, 22(1):258.

Luecken, M. D. and Theis, F. J. (2019). Current best practices in single-cell RNA-seq analysis: a tutorial. *Molecular Systems Biology*, 15(6):e8746.

Luecken, M. D., Büttner, M., Chaichoompu, K., et al. (2022). Benchmarking atlas-level data integration in single-cell genomics. *Nature Methods*, 19(1):41–50.

Margolin, A. A., Nemenman, I., Basso, K., et al. (2006). ARACNE: an algorithm for the reconstruction of gene regulatory networks in a mammalian cellular context. *BMC Bioinformatics*, 7(Suppl 1):S7.

Moerman, T., Aibar Santos, S., Bravo González-Blas, C., et al. (2019). GRNBoost2 and Arboreto: efficient and scalable inference of gene regulatory networks. *Bioinformatics*, 35(12):2159–2161.

Pedregosa, F., Varoquaux, G., Gramfort, A., et al. (2011). Scikit-learn: machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830.

Polański, K., Young, M. D., Miao, Z., Meyer, K. B., Teichmann, S. A., and Park, J.-E. (2020). BBKNN: fast batch alignment of single cell transcriptomes. *Bioinformatics*, 36(3):964–965.

Pratapa, A., Jalihal, A. P., Law, J. N., Bharadwaj, A., and Murali, T. M. (2020). Benchmarking algorithms for gene regulatory network inference from single-cell transcriptomic data. *Nature Methods*, 17(2):147–154.

Replogle, J. M., Saunders, R. A., Pogson, A. N., et al. (2022). Mapping information-rich genotype–phenotype landscapes with genome-scale Perturb-seq. *Cell*, 185(14):2559–2575.e28.

Squair, J. W., Gautier, M., Kathe, C., et al. (2021). Confronting false discoveries in single-cell differential expression. *Nature Communications*, 12(1):5692.

Stuart, T., Butler, A., Hoffman, P., et al. (2019). Comprehensive integration of single-cell data. *Cell*, 177(7):1888–1902.e21.

Svensson, V. (2020). Droplet scRNA-seq is not zero-inflated. *Nature Biotechnology*, 38(2):147–150.

The Tabula Sapiens Consortium. (2022). The Tabula Sapiens: a multiple-organ, single-cell transcriptomic atlas of humans. *Science*, 376(6594):eabl4896.

Theodoris, C. V., Xiao, L., Chopra, A., et al. (2023). Transfer learning enables predictions in network biology. *Nature*, 618(7965):616–624.

Traag, V. A., Waltman, L., and van Eck, N. J. (2019). From Louvain to Leiden: guaranteeing well-connected communities. *Scientific Reports*, 9(1):5233.

Tran, H. T. N., Ang, K. S., Chevrier, M., et al. (2020). A benchmark of batch-effect correction methods for single-cell RNA sequencing data. *Genome Biology*, 21(1):12.

Travaglini, K. J., Nabhan, A. N., Penland, L., et al. (2020). A molecular cell atlas of the human lung from single-cell RNA sequencing. *Nature*, 587(7835):619–625.

Wolf, F. A., Angerer, P., and Theis, F. J. (2018). SCANPY: large-scale single-cell gene expression data analysis. *Genome Biology*, 19(1):15.